DA18 Does your total match the true total? Clean, merge and reconcile

Week 6, Messy data and two tables | Session 18 of 24 | Python for Data Analysis

Today's question. Which category earns the most in the cleaned data, and by how much do our totals fall short of the truth?

Objective. By the end: you can clean a raw export, merge it with a lookup table, and report a total together with its named difference from the true total.

> WHERE THIS IS USED
> Accountants and auditors compare two totals and explain the gap. Banks and tax teams do this every month. Can you explain why your total is a little short of the true one?

Before this session: DA17 Merging two tables and the fan-out trap. If you cannot explain the last example, use its error card with the instructor first.

Make

In DA18.ipynb start from sunrise_sales_raw.csv. 1) Clean it with a function and report the rows kept, rows excluded and the cleaning steps. 2) Merge the clean table with the products table on product_id. Check that the key is unique and that the row count is unchanged. 3) Total the revenue by category. 4) Compare the cleaned total with the true total in sunrise_sales.csv. Name the residual and its cause. 5) Write two limitations. Run from a restarted kernel, save DA18.ipynb with a short explanation of your checks, and share them by the route in START_HERE.pdf.

Save your work as DA18. Keep the worked example and put your own attempt below it. Write two sentences: what you changed and how you checked it.

Error card

You see	It means	Fix
The category totals add to 6375900, not 6250500	The clean table was not used, or the excluded rows were put back by a merge.	Total from the cleaned table and compare with len() at each step.
MergeError or a larger row count after the merge	The key is repeated in the products table.	Check is_unique on the lookup table before you merge.

Exit ticket

I can show my steps, my checks and the named reason my total is not the true total.

AI use follows AI_USE_POLICY.pdf. In weeks 1 to 6 no AI-generated solutions are used for exercises, labs or build tasks. Practise for 20 to 30 minutes between classes by repeating the Make task with one value changed. If you are stuck for 30 minutes, save the error and bring it to class. A file saved on your laptop has not been shared until you use the route in START_HERE.pdf.